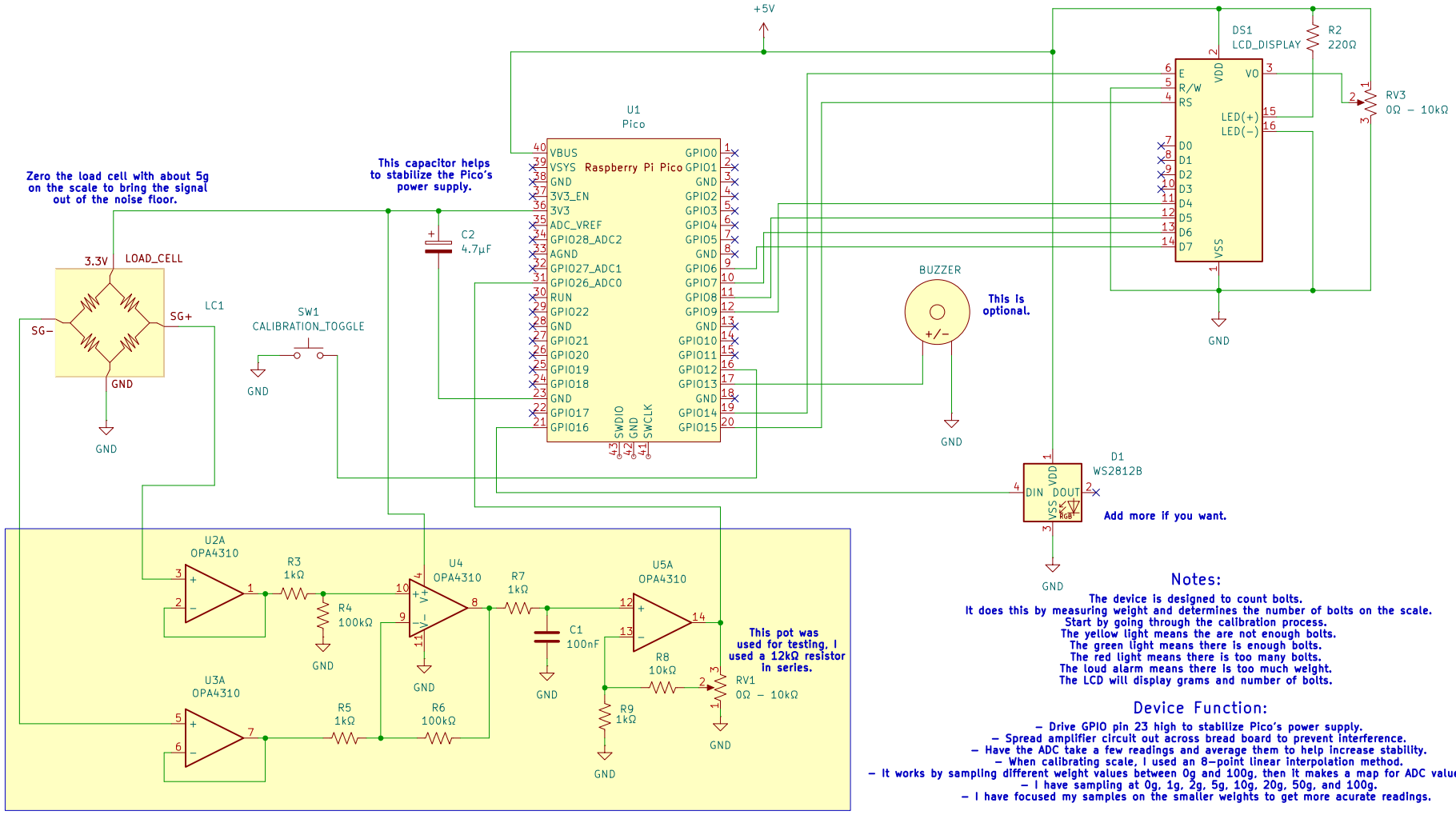


Signal Amplifier Circuit

(all opamps are part of the IC OPA4310)

Notes:

The device is designed to count bolts.
It does this by measuring weight and determines the number of bolts on the scale.
Start by going through the calibration process.
The yellow light means the are not enough bolts.
The green light means there is enough bolts.
The red light means there is too many bolts.
The loud alarm means there is too much weight.
The LCD will display grams and number of bolts.

Device Function:

– Drive GPIO pin 23 high to stabilize Pico's power supply.
– Spread amplifier circuit out across bread board to prevent interference.
– Have the ADC take a few readings and average them to help increase stability.
– When calibrating scale, I used an 8–point linear interpolation method.
– It works by sampling different weight values between 0g and 100g, then it makes a map for ADC values to Grams.
– I have sampling at 0g, 1g, 2g, 5g, 10g, 20g, 50g, and 100g.
– I have focused my samples on the smaller weights to get more acurate readings.